AUGUST/RESIT EXAMINATIONS 2021/2022

MODULE: CA119 - Data Science and Databases

PROGRAMME(S):

DS	BSc in Data Science
AP	BSc in Applied Physics
ECSAO	Study Abroad (Engineering & Computing)

YEAR OF STUDY: 1,3,O

EXAMINER(S):

Mark Roantree	(Internal)	(Ext:5636)
Assoc. Prof. Sheila McBreen	(External)	External
Prof. Gerard O'Connor	(External)	External

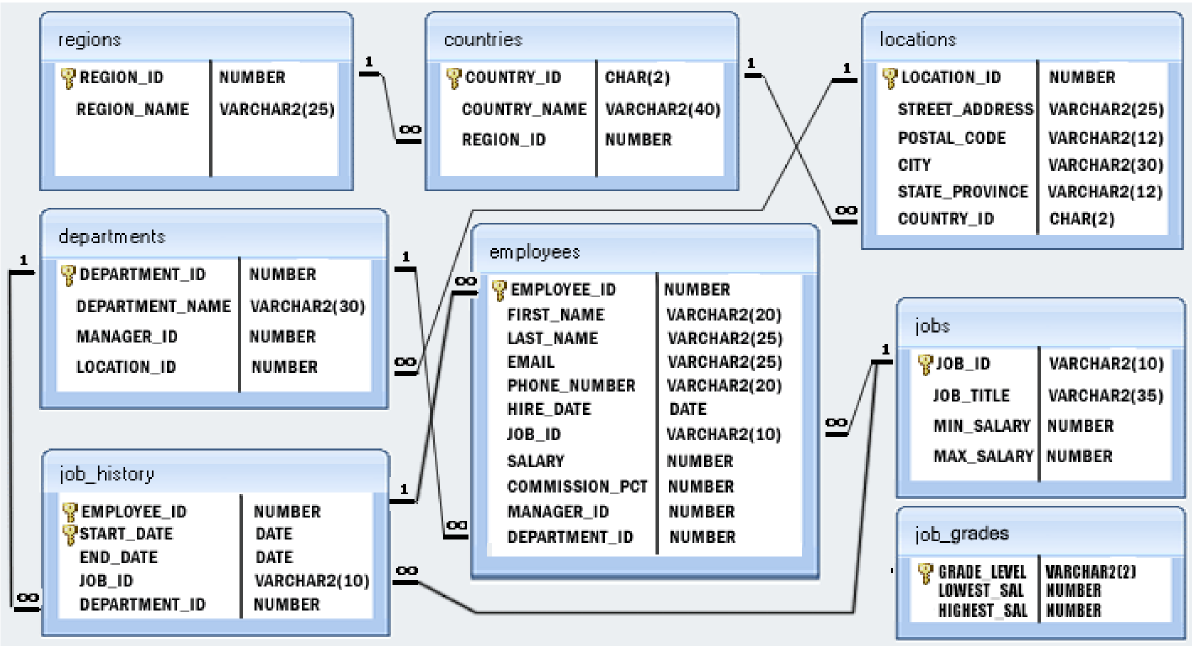
TIME ALLOWED: 2 Hours

INSTRUCTIONS: **Answer 4 questions. All questions carry equal marks.**

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.
The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

QUESTION 1 (SQL Programming) [TOTAL MARKS: 25]



HR Department Schema

Q 1(a) *[4 Marks]*

Display the name, hire date, salary, and department number for those employees whose first name does not contain the letter T.
Sort the results in ascending order by department number.

Q 1(b) *[4 Marks]*

Display the name of all employees and the name of their manager.

Q 1(c) *[6 Marks]*

Display the employee number, name, job ID and salary for all employees whose salary is smaller than any salary of those employees whose job title is MK_MAN.
Exclude employees who have the Job ID ‘MK_MAN’ from your results.

Q 1(d) *[5 Marks]*

Display the department name, average salary and number of employees working in their department who received a commission.

Q 1(e) *[6 Marks]*

Display job ID, number of employees, sum of salary, and difference between highest salary and lowest salary for each job.

[End of Question 1]

QUESTION 2 (The Relational Model) [TOTAL MARKS: 25]

SID	Name	Address	CourseID	CourseName	TutorID	TutorName
20181011	Michael Murphy	Dundalk	DC101	Computer Applications	750001	Jack Sharp
20181015	Ann Burke	Dunboyne	DC101	Computer Applications	750003	Pauline Burke
20181022	Tom Flynn	Finglas	DC101	Computer Applications	750004	Nora Martin
20181111	Mary Burke	Glasnevin	DC101	Computer Applications	750004	Nora Martin
20181211	Brian Byrne	Glasnevin	DC102	Enterprise Computing	750004	Nora Martin
20181256	Ellen Molloy	Glasnevin	DC102	Enterprise Computing	750007	Eilis Flynn
20181260	Deirdre flynn	Finglas	DC101	Computer Applications	750007	Eilis Flynn
20181275	Alan Marr	Finglas	DC102	Enterprise Computing	750001	Jack Sharp

Relation R (STUDENTS)

Q 2(a) *[5 Marks]*

- What is the cardinality of the STUDENTS relation above?
- What is the degree of this relation?
- Which of the above is related to the intension of the relation? Explain why you gave this answer.

Q 2(b) *[8 Marks]*

- Write the formula for the cartesian product of the 2 tables *R* in this Question and table *S* in Question 3.
- Write the result of this expression to your exam paper.

Q 2(c) *[5 Marks]*

Define what is meant by a *Domain* constraint. Provide an example of breaking this constraint for the schema in Relation R.

Q 2(d) *[4 Marks]*

List what you believe to be all superkeys for the STUDENTS relation.

Q 2(e) *[3 Marks]*

Using the relation STUDENTS, provide an example of a violation of the *delete constraint*. Explain how the violation occurs.

[End of Question 2]

QUESTION 3 (Functional Dependencies) **[TOTAL MARKS: 25]**

CourseID	Department	Head
DC101	Department of Computing	Prof. Murphy
DC102	Department of Biology	Prof. Swan
DC103	Department of Physics	Prof. McDonald

Relation S *(*PROFESSORS)

Q 3(a) *[10 Marks]*
Assuming that a functional dependency is of the form L→ R where L represents one or more (determinant) attributes and R a different set of attributes, list 5 functional dependencies for the STUDENTS relation in Question 2.
Provide a *different* value for L in each dependency.
(Hint: there are many possibilities).

Q 3(b) *[4 Marks]*
Provide an example of an *insertion anomaly* for the relation STUDENTS (Question 2) and the relation PROFESSORS above.

Q 3(c) *[6 Marks]*
What is meant by the *dependency preservation* property? In your answer, explain the circumstances in which it may occur and how it should be resolved.

Q 3(d) *[5 Marks]*
Assume you have a relation R with attributes {a,b,c,d,e}. Briefly describe an algorithm to locate functional dependencies in R.

[End of Question 3]

QUESTION 4 (Normalisation) **[TOTAL MARKS: 25]**

Q 4(a) *[6 Marks]*

- In what normal form would you consider the relation STUDENTS (see Question 2) to be? Why?
- In what normal form would you consider the relation PROFESSORS (see Question 3) to be? Why?

Q 4(b) *[13 Marks]*
Taking the relation STUDENTS, explain the steps you would take to bring this relation to third normal form. At each step, highlight the specific functional dependency that controlled your decomposition.

Q 4(c) *[6 Marks]*
Take the relation PROFESSORS to create an unnormalised form of the relation to your answer book. Explain what you did and why it became an unnormalised relation.

[End of Question 4]

QUESTION 5 (Entity Relationship Modelling) ***[TOTAL MARKS: 25]***

You must design a database for a Sports Centre in Dublin which has a gymnasium, swimming pool, basketball and squash courts. Thus, the Sports Centre operates four types of Activity. It also has 3 different membership types: pool only, gym only, or all 4 activities.

Q 5(a) **[10 Marks]**
Create an Entity-Relationship diagram for this application where each entity has a minimum of 4 attributes. Each entity should have at least one relationship so that all entities are connected to at least one other entity. Use different types of relationships in your diagram. State clearly any assumptions that you make.

Q 5(b) **[7 Marks]**
What type of relationship connects Member and Activity? Why is this type of relationship necessary in your design? When transforming from an E-R model to a relational schema, how would you manage this relationship

Q 5(c) **[8 Marks]**
What is meant by a participation constraint? Provide a description of both types of participation constraint. In your answer provide an example from the gymnasium case study of each type of participation constraint.

[End of Question 5]

[END OF EXAM]